PRODUCT REQUIREMENT DOCUMENT (PRD)

FluxClip Studio — UI Design & Pipeline Logic Specification

Scope: Screen 1 (Uploads Tab) & Screen 2 (Clips Tab) • **Architecture:** Jetpack Compose UI + OpenShorts Pipeline • **Release:** Production v2.0

1. Screen 1: Uploads Tab (Ingestion & Pipeline Trigger Logic) SCREEN 1

Acts as the primary media ingestion gateway for FluxClip. Handles source video payload validation and hands off execution tasks to the OpenShorts pipeline daemon.

UI Component	Jetpack Compose Target	Pipeline Logic & Behavioral Rules
Remote URL Field	`OutlinedTextField` (56dp)	Accepts YouTube/streaming links; passes URL to native `yt-dlp` backend engine to stream audio and video streams.
Local File Card	`Card` / `Surface`	Invokes Android system picker for local MP4/MOV/MKV/WebM files up to 2GB. Uses `FFmpeg` for container demuxing and keyframe indexation.
Generate Clips CTA	`Button` (Gradient Fill)	Validates input; triggers pipeline processing (audio separation, Whisper transcription, and moment ranking). Transitions state upon daemon ack.
Navigation Bar	`NavigationBar` (56dp)	Bottom bar with **[Uploads] (Active)** and **[Clips] (Inactive)** tabs.

2. Screen 2: Clips Tab (Review & Export Logic) SCREEN 2

Renders segments produced by the OpenShorts processing pipeline. Displays ranked viral clips, presents render playback for the selected clip, exposes framing/caption metadata, and provides single-clip artifact export.

2.1. UI Layout & Visual Components

UI Component	Jetpack Compose Target	Design & Layout Specifications
Title Header	`Text`	Clean serif headline: *"Generated Clips"* (35sp, no decorative magic wand elements).
Clip Carousel	`LazyRow`	Single-row horizontal scrollable strip showing thumbnail cards with duration badges (*Clip 1, Clip 2, Clip 3...*).
Video Display	`AndroidView(PlayerView)`	Large vertical 9:16 aspect ratio player displaying the backend-rendered segment for the selected clip.
Segment Details	`Text`	Specifies segment metadata: *"Clip [Index] • 9:16 vertical reframing / [Duration]s"*.
Single Export CTA	`Button`	Primary action button: *"Export Clip ↓"* (styled with downward-pointing download arrow).
Navigation Bar	`NavigationBar` (56dp)	Bottom bar with **[Clips] (Active)** and **[Uploads] (Inactive)** tabs.

2.2. Functional Logic & OpenShorts Pipeline Mapping

- **Segment Ingestion & Discovery:** Binds to artifacts in pipeline output folder. Each carousel item maps to a segmented moment with start/end timestamps, virality score, and pre-extracted thumbnail frame.
- **Vertical Reframe & Caption Alignment:** Displays 9:16 vertical crop generated by backend speaker-tracking logic. Subtitles aligned via Transformers.js/Whisper are embedded into the rendered stream.
- **Single Clip Export Action:** Triggering *"Export Clip ↓"* writes selected segment MP4 from app staging to device public storage via Android `MediaStore.Video` API.

3. Boundary Conditions & Excluded Non-Native Features

Ensures strict alignment with the upstream OpenShorts repository and eliminates logic collisions:

- **No Batch ZIP Export:** Multi-clip zip compression and batch packaging do not exist in the native pipeline; Screen 2 supports strictly discrete single-clip export (`Export Clip ↓`).
- **No Custom Interactive Scrubber State:** Clips are pre-rendered MP4 artifacts with burned-in subtitles and reframing; client-side interactive re-timing and dynamic crop manipulation are excluded.
- **Decoupled Thread Architecture:** All file I/O and process polling execute in background coroutines, keeping the Jetpack Compose UI layer purely reactive.